Mini project 2026 - Modelling a 3D printer

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The mini project is there to tie the lessons learned from the course together and simulate a complex system in a holistic way. This year, you'll build a simulation of a 3D printer. This is a group project, make sure you collaborate and spread the workload.

Deliverables

We expect you to provide the following results:

1. A report of **no more than 20 pages** presenting your results. We expect clean plots, complete derivations and thorough explanations of all your choices. Please mark clearly the answers to each of the questions with their corresponding question number.
2. The code used to achieve your results. You are free to choose a programming language of your liking. We suggest Python, Julia or Matlab.
3. A code repository, tracking your progress. We expect you to create the repository **at the start** of your project and to continuously maintain it during the project. Please keep the repository **private** during the mini project. When you make your final submission on Brightspace please indicate how we can access your repository. You can either publish your repository and provide the link in the report, or add one of the supervisors to your repository. If you choose the latter option, please reach out for our account details on the platform you are working on.

Even though it's not part of this course, try to practice good "git hygiene" by committing, branching, merging. There are many tutorials introducing you to GitHub, GitLab, etc. Also consider that you can use this project as a showcase for applications after your studies.

For inspiration on how to structure the report, which information to provide and how, have a look at solutions of previous years mini-projects on Brightspace. If you have questions, reach out during the Q&As, on the Brightspace discussion board, or via MS Teams.

AI usage

Make sure to read and implement the TU Delft AI guidelines available here. When writing the acknowledgment about the (eventual) use of AI, please use the standard version of it.

Furthermore, the use of Generative AI tools (e.g., Copilot) is restricted/prohibited in this course for the following activities and/or assessments:

- Generating design calculations or schematic drawings without performing the underlying calculations or design work yourself.
- Creating presentation slides, posters, reflection pieces, professional development statements, or reports intended to demonstrate your own experiences.
- Generating specific sections of assignments, essays, reports, or other coursework.
- Producing data analysis outputs.
- Producing or creating entire sections of code for submission.

In case there is suspicion of violation of this requirements, students will be summoned for a meeting with the instructors in order to explain how they did adhere to these rules and provide further proof that results of the assignments were generated according to the rules.

Introduction

3D Printers have the basic task of extruding filament at a given x, y, z coordinate at time t. The mechanism that achieves that movement determines the machine's costs, print quality, and robustness. Out of many different options, the Core XY mechanism is a popular choice for modern printers. Different than Cartesian designs, where one

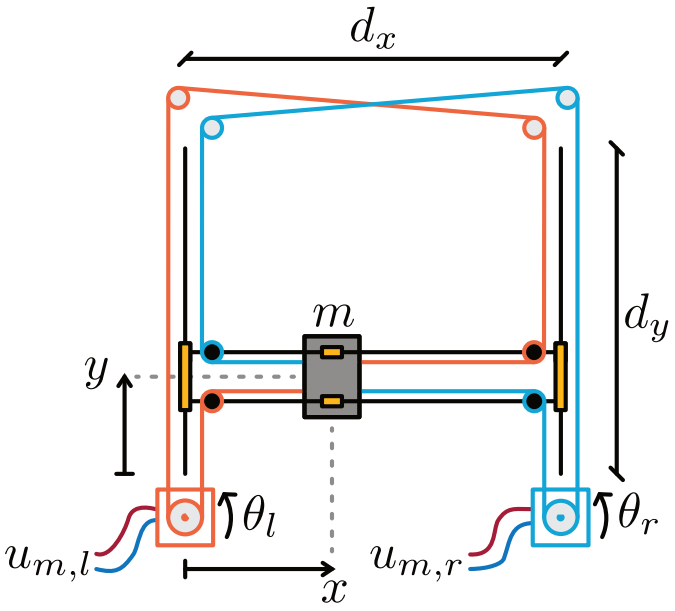


Figure 1: Core XY printer mechanism excluding the Z axis. The belts are blue and orange, and the driving motors are coloured respectively. Linear bearings are marked in yellow and slide on rails (black). The tool head with mass m is coloured grey. The circles, with the exception of the motors, mark bearings. Bearings with a black colour are fixed to the rail below. The electrical input to the motors is shown as red and blue wires.

motor drives one axis, here each motor simultaneously changes the x and y position of the tool head. This results in a more complex kinematic model, but allows a more sturdy casing and faster printing speeds. Your task is to model and simulate the system.

VARIABLES AND PARAMETERS Most of the variables and parameters in this problem are defined in the table below. Note that the introduction of more variables and parameters may be necessary. If needed, choose sensible values for parameters and variables yourself.

Name	Symbol
Motor angle left	θ_l
Motor voltage left	$u_{m,l}$
Motor current left	$i_{m,l}$
Motor radius	r_m
Motor induction	L_m
Motor resistor	R_m
Motor inertia	J_m
Motor constant	K_m
Tool head position	x, y
Tool head mass	m
Nozzle voltage	u_n
Nozzle current	i_n
Nozzle capacitor	C_n
Nozzle thermal capacity	$C_{t,n}$
Nozzle thermal resistance	$R_{t,n}$
Nozzle resistor	R_n
Nozzle temperature	T_n
Environmental temperature	T_e
Bearing friction rolling	μ_r
Bearing friction linear x-direction	$\mu_{l,x}$
Bearing friction linear y-direction	$\mu_{l,y}$
Print bed temperature	T_b
Print area dimension x	d_x
Print area dimension y	d_y

Table 1: Variables used in the mini project. The right motor variables are analogous to the left motor variables. You may need to define additional variables. If needed, choose sensible values for parameters and variables yourself.

1 Physical Modelling

Q1.1 Kinematics (1 pt.)

Model the kinematics of the system. At the end you should have an expression of the x, y coordinates and velocities of the print head based on the motor angles θ_l and θ_r. We expect formulations for the variables:

$$x(\theta_l, \theta_r),\ y(\theta_l, \theta_r) \tag{1}$$

$$\dot{x}(\dot{\theta}_l, \dot{\theta}_r),\ \dot{y}(\dot{\theta}_l, \dot{\theta}_r). \tag{2}$$

Q1.2 Dynamics (2 pts.)

Model the dynamics of the system. At the end of this part you should have a state-space description of the system with the voltages of the motors as inputs of the form

$$\dot{\boldsymbol{x}} = f(\boldsymbol{x}, \boldsymbol{u}) \tag{3}$$

with $\boldsymbol{x} = [\theta_l, \theta_r, \cdots]^\top$ and $\boldsymbol{u} = [u_l, u_r]^\top$.

An important caveat! Your colleague ordered the wrong motors, so instead of the typical stepper motors, you only have voltage-controlled DC motors at your disposal. As a result, you have to consider the following circuit to model the motor:

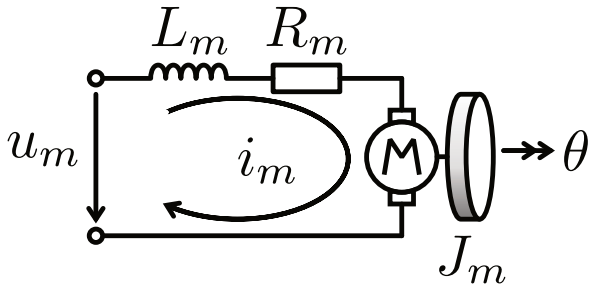


Figure 2: The DC motor is modelled as an LR circuit, coupled to an electrical-to-rotational mechanical transducer. The used variables are L_m : inductor, R_m : resistance, i_m : current, u_m : input voltage, J_m : motor inertia, and θ : motor angle. We neglect friction and the shaft is considered rigid.

Q1.3 Decouple the x and y dynamics (1 pt.)

Investigate how you can transform the state-space model to decouple the x and y motions. You should arrive at two decoupled state-space descriptions:

$$\dot{\boldsymbol{x}}_x = f_x(\boldsymbol{x}_x, \boldsymbol{u}_x)\,, \tag{4}$$

$$\dot{\boldsymbol{x}}_y = f_y(\boldsymbol{x}_y, \boldsymbol{u}_y)\,,$$

with $\boldsymbol{x}_x = [x, \cdots]^\top$, $\boldsymbol{x}_y = [y, \cdots]^\top$, and $\boldsymbol{u}_x$, $\boldsymbol{u}_y$ being functions of u_l and u_r.

Hint: *Review your solution from Q1.1.*

Q1.4 *Drawing shapes (1 pt.)*

Design input signals such that the tool head follows an elliptic path. Draw another closed shape of you choosing (i.e. a circle, a symbol from poker cards, ... use your imagination and be original!).

The *Lissajou curves* can be one source of inspiration. Are shapes with sharp corners more difficult to draw than ones with rounded corners?

Show the working of your model through plots!

Q1.5 *Model the Nozzle (1 pt.)*

The nozzle is heated by a resistor and is subject to heat exchange with the environment and the filament. Input to the circuit is a controllable voltage. Model the governing differential equation(s). At the

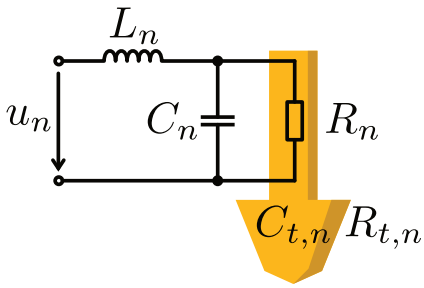


Figure 3: Nozzle circuit with the inductor L_n, the resistor R_n, the capacitor C_n, as well as the thermal capacity and resistance of the nozzle $C_{t,n}$ and $R_{t,n}$.

end, you should have a state-space discription of the form

$$\dot{\boldsymbol{x}}_n = f_n(\boldsymbol{x}_n, u_n) \tag{5}$$

with $\boldsymbol{x} = [T_n, \cdots]^\top$. You may combine the thermal loss to the environment and the filament into one loss with respect to the environmental temperature T_e, which you can choose.

Q1.6 *Printing a circle of filament (PDE) (2 pts.)*

When printing a 3D object layer-by-layer, care must be taken to manage the temperature of the solidifying filament already deposited, ensuring it is optimal before printing the next layer on top.

In the following figure, you see the printing of the first layer of a cylinder having an elliptical cross-section. The filament is deposited onto a heated base plate to ensure it does not cool too quickly.

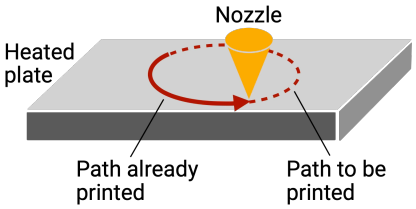


Figure 4: Printing one layer of an object whose cross-section is a closed shape, in this case an ellipse.

Develop a 1D distributed-parameter model of the filament cooling down, and solve it numerically to show how long it takes for it to cool. This information is useful for planning the print of the next layer.

When modelling, you can assume for simplicity a square cross section of the printed filament. The lower side of it is in contact with the heated plate, while the other sides are exposed to the air at ambient temperature. The simulation should start as soon as the filament is being printed, which means your computational domain is growing over time. When the entire path has been printed, no more filament is deposited, but the simulation should continue until the temperature of the printed filament is nearly uniform.

To model the geometry, you can take inspiration from the closed circuit cooling pipe of the previous year's mini project. But be careful: in the present case, the computational domain changes over time!

2 *Finite State Machine*

Q2.1 *Finite State Machines - Filament replacement (2 pts.)*

Next, you will model some basic high-level control operations of your 3D printer. Namely, you will model the replacement of filament in these two situations:

- When a new colour is required to be printed (C), or
- When the currently being used spool is finished (F).

We start modelling these two distinct situations as being triggered by two, independent, discrete signals C and F. Once either of these events is detected, the following set of steps needs to be taken:

1. Printing stops, saving the current location of the print head.
2. Moving to a wipe-tower location (x_p, y_p), which we assume to be at $(0, 0)$.
3. Excess filament is wiped (by printing on the wiping location).
4. The nozzle heat is switched off.
5. New filament is loaded.
6. The nozzle heat is switched on.
7. Printing is resumed.

Model the operation first as a Finite State Machine (FSM) of the 3D printer, including a state for “printing" and any necessary states to cover the filament replacement process.

Q2.2 *Automata - Parked detection (3 pts.)*

Your colleague forgot to order end-of-line sensors, but thankfully, the motors include rotational sensors you can use as a virtual end-of-line sensor. Simply put, you can detect an end-of-line by testing whether or not the motors are rotating to produce a vertical or horizontal motion when the corresponding voltages are applied to them. Let us assume the printer head is always brought to the origin via the concatenation of two types of motions:

- Vertical (V) until we hit the bottom of the printing area,
- Horizontal (H) until we hit the left-most side of the printing area.

Furthermore, consider that your design includes a system that periodically checks the value of the voltages applied (to determine the type of motion commanded), and the angular velocity sensors of the motors.

Consider signal sequences $\sigma : \mathbb{N} \to M \times m$ where $M = \{H, V\}$, and $m = \{+, 0\}$. A pair (M_i, m_i) denotes “type of motion" (horizontal, vertical), and “angular velocity" (positive or zero). As an example, one such sequence could be $\sigma = (H,+)(V,+)(H,+)(H,0)(V,+)(V,0)$.

Construct an automaton that could be used as a *head-parked detector* by accepting the sequences σ that lead to the printer head being correctly parked. Note that the same automaton with minor adjustments can be used to detect any of the other three corners of the printing area.

3 Hybrid system

Finally, you need to put together a complete simulation model of your 3D printer. To that end, you need to connect your physical model from Q1 with the models from Q2 employing some event signals. You may consider in your model the following events, and any additional ones you may deem necessary:

1. New colour is required.
2. Filament finished.
3. Excess filament removed.
4. Nozzle temperature is high.
5. Nozzle temperature is low.
6. Motion commands V and H.
7. Angular motion.

***Q3.1** - Filament replacement (2 pts.)*

Follow the implementation guidelines and tips from the lecture!

Build a hybrid system accounting for the physics of Q1 and the FSM designed on Q2. Simulate the resulting system and provide illustrative plots.

Make sure that your description clearly contains:

- Proper definitions of the discrete events relating to the state variables of your physical model.
- All the necessary invariants, guards and reset maps.

Hints:

- *Collect all state-space descriptions into one model of the form $\dot{\boldsymbol{x}} = f(\boldsymbol{x}, \boldsymbol{u})$ and implement it as one function `dx=sys(x,u)` which you can pass on to a solver, along with the event triggers.*
- *Of the events defined above, the first three may be considered as purely discrete events triggered by some sensor. An alert designer may notice that signals 2 and 3 could potentially be detected as a function of the distance printed, or the effective printing time.*

***Q3.2** - Draw a square (1 pt.)*

Extend the hybrid system to draw a (perfect) square figure using only the angular motion sensors on the DC motors. Simulate this system and provide illustrative plots.